

POLYTECHNIQUE MONTRÉAL

affiliée à l'Université de Montréal

Numerical Modeling of Multilayer Aircraft Ice Accretion

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DEDICATION

*À tous mes amis du labos,
vous me manquerez. . .*

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Texte / Text.

RÉSUMÉ

Le résumé est un bref exposé du sujet traité, des objectifs visés, des hypothèses émises, des méthodes expérimentales utilisées et de l'analyse des résultats obtenus. On y présente également les principales conclusions de la recherche ainsi que ses applications éventuelles. En général, un résumé ne dépasse pas trois pages.

Le résumé doit donner une idée exacte du contenu du mémoire ou de la thèse. Ce ne peut pas être une simple énumération des parties du manuscrit. Le but est de présenter de façon précise et concise la nature, l'envergure de la recherche, les sujets traités, les questions de recherche ou les hypothèses soulevées, les méthodes utilisées, les principaux résultats ainsi que les conclusions retenues. Un résumé ne doit jamais comporter de références ou de figures.

ABSTRACT

Written in English, the abstract is a brief summary similar to the previous section (Résumé). However, this section is not a word for word translation of the abstract in French.

The abstract is a brief statement of the subject matter, objectives, research questions or hypotheses, experimental methods and analysis of results. It also presents the main research conclusions and their possible applications. In general, an abstract should not exceed three pages.

The abstract should provide an exact idea of the thesis or dissertation's contents and it cannot be a simple enumeration of the manuscript's parts. The goal is to precisely and concisely present the nature and scope of the research. An abstract should never include references or figures. If the thesis or the dissertation is in English, the résumé (French-language abstract) should come first followed by the abstract.

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LIST OF SYMBOLS AND ACRONYMS

IETF	Internet Engineering Task Force
OSI	Open Systems Interconnection

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CHAPTER 1 INTRODUCTION

This chapter introduces the context of aircraft icing, its impact on flight safety, and its role in certification. It reviews the main ice morphologies, regulatory requirements, and certification tools, before presenting the challenges of numerical ice accretion modeling. The chapter concludes by defining the thesis objectives and outlining the organization of the thesis.

1.1 Context

In the context of climate change and increasingly stringent environmental regulations, the aeronautical industry is moving toward the development of more efficient and lower-emission aircraft. This transition introduces new aerodynamic, structural, and operational challenges, while maintaining flight safety as the primary requirement. It also increases the need for reliable certification procedures capable of assessing aircraft performance and safety over a wide range of operating conditions.

Among the phenomena considered during aircraft certification, in-flight icing remains one of the major concerns. Ice accretion occurs when an aircraft flies through clouds containing supercooled water droplets¹. When these droplets impinge on exposed surfaces and the local thermal conditions allow freezing, ice can accumulate on the aircraft geometry. The resulting ice shapes may significantly modify the aerodynamic configuration of the aircraft, leading to increased drag, reduced lift, altered stall behavior, and degraded controllability [10–12]. While aircraft icing is commonly associated with large exposed components such as wings, empennages, the aircraft nose, and engines, it may also affect smaller but critical components, including probes and sensors, as illustrated in Figure 1.1. Ice accretion on these components can produce erroneous measurements and, consequently, compromise flight safety.

Aircraft icing was not initially regarded as a primary concern during the early years of powered flight. Because aircraft were not yet equipped for routine instrument flight, pilots generally avoided clouds, and encounters with icing conditions were mostly inadvertent. This situation changed in the mid-1920s, when the U.S. Air Mail Service began operating scheduled day-and-night routes between New York and Chicago. These operations exposed pilots to icing conditions on a regular basis, and several serious incidents and accidents in 1926 and 1927 highlighted the need for practical solutions to the icing problem [10]. Early

¹Supercooled water droplets are liquid water droplets that remain unfrozen despite being at temperatures below the freezing point.



(a) Aircraft engine [13]



(b) Pitot probe [14]

Figure 1.1 Examples of ice accretion on aircraft components.

research efforts were therefore mainly directed toward ice-protection systems (IPS)², including heated aircraft components, waterproof coatings, greased propellers, electrically heated pitot-static tubes, and pneumatic de-icing boots. Although some of these solutions proved useful in specific applications, they remained largely empirical and did not provide a complete understanding of the physical mechanisms governing ice accretion.

Over the following decades, repeated claims that aircraft icing had been effectively solved proved premature, as operational experience continued to show that the phenomenon required dedicated investigation under controlled experimental conditions. This need motivated the development of controlled icing research capabilities, culminating in the construction of the Icing Research Tunnel (IRT) in Cleveland in the 1940s [10]. The IRT enabled systematic investigations of supercooled cloud conditions, droplet impingement, ice-accretion processes, aerodynamic performance degradation, and the performance of anti-icing and de-icing systems. These experimental studies played a central role in improving the physical understanding of aircraft icing and later provided essential data for the development, validation, and certification of numerical icing prediction methods [15].

1.2 Ice Morphologies

Depending on the atmospheric and aerodynamic conditions, different ice morphologies may develop on aircraft surfaces. At low temperatures, rime ice forms when supercooled droplets freeze almost immediately upon impact. This rapid solidification produces a relatively smooth, streamlined, and opaque ice shape. The opaque appearance of rime ice is asso-

²Ice protection systems were divided into two categories : Anti-icing systems are designed to prevent ice from forming on protected surfaces, whereas de-icing systems allow ice to form temporarily before removing it.

ciated with air trapped within the ice during the rapid freezing process.

As the temperature increases toward the freezing point, glaze ice may form. In this regime, the impinging droplets do not freeze completely upon impact. Instead, part of the liquid water remains on the surface, forming a thin water film that can run back before freezing downstream. Glaze ice is therefore usually denser and more transparent than rime ice, and it often leads to more complex ice shapes, including horn-like structures, feathers, and, on swept configurations, three-dimensional features such as scallops or lobster tails (Fig. 1.2).

A third regime, mixed ice, occurs as a transition between rime and glaze conditions. In this case, glaze-like ice may form near the stagnation region, where the impingement rate is high and the water does not freeze completely on impact. Farther downstream, the remaining droplets or runback water may freeze more rapidly, producing rime-like deposits. Mixed ice therefore combines features of both rime and glaze ice.



Figure 1.2 Ice accretion on the leading edge of the CRM hybrid wing model tested in the NASA Icing Research Tunnel. Photos taken from [1]

1.3 Regulations & Certification

The diversity of ice morphologies and the severity of their aerodynamic effects make in-flight icing a critical consideration in aircraft certification. Regulatory authorities require aircraft to demonstrate safe operation within prescribed icing environments, which are defined

through envelopes of atmospheric and flight conditions. A major international reference for the certification of transport-category airplanes is Title 14 of the Code of Federal Regulations, Part 25, adopted by the Federal Aviation Administration (FAA). Within this framework, Appendix C has historically defined the reference atmospheric icing envelopes used for the design and certification of IPS. These envelopes, in use since 1964, describe continuous maximum and intermittent maximum icing conditions in terms of parameters such as liquid water content, temperature, altitude, cloud horizontal extent, and representative droplet diameter [16]. They were primarily established for conventional cloud icing conditions, where the droplet median volume diameter (MVD) is typically limited to the range of standard supercooled droplets (SSD), up to approximately $50\text{ }\mu\text{m}$.

However, several in-service events demonstrated that aircraft could also encounter icing environments outside the original Appendix C envelopes. A major example is the crash of American Eagle Flight 4184, an ATR-72, near Roselawn, Indiana, on October 31, 1994. The accident investigation identified icing as a central factor in the loss of control and highlighted the safety implications of supercooled large droplet (SLD) conditions, including freezing drizzle and freezing rain. These environments may contain droplets significantly larger than those considered in Appendix C, with droplet diameters reaching several hundred micrometres, typically up to about $500\text{ }\mu\text{m}$ in freezing-drizzle conditions, and up to approximately $1000\text{ }\mu\text{m}$ in freezing-rain conditions.

This recognition led to additional regulatory developments, culminating in the introduction of Appendix O to 14 CFR Part 25 for SLD icing conditions. Appendix O extends the certification envelope to include freezing drizzle and freezing rain, thereby addressing icing scenarios in which large droplets may splash, rebound, or break up before freezing. These mechanisms may lead to ice formation downstream of the protected region. Consequently, aircraft seeking certification for flight in icing conditions must demonstrate safe operation under these conditions.

The certification of IPS may rely on several complementary approaches, including flight testing, icing wind-tunnel testing, and numerical simulation. Flight tests provide the most representative assessment of aircraft behavior in realistic operating conditions and may be conducted either in natural icing environments or using artificial ice shapes. However, such tests are expensive, difficult to reproduce, and may involve safety risks, particularly when degraded aerodynamic performance or controllability must be evaluated.

Icing wind tunnels therefore became an essential complementary experimental tool for studying ice accretion and evaluating IPS performance under controlled conditions. Facilities such as the NASA IRT, the National Research Council (NRC) Low-Speed Icing Wind Tunnel in

Canada [17], the Rail Tec Arsenal (RTA) facility in Austria [18], and the Office National d'Études et de Recherches Aérospatiales (ONERA) icing facilities [19] in France have contributed extensively to the development of experimental databases for icing research and certification support. Compared with flight testing, wind-tunnel experiments offer better control of atmospheric parameters, and detailed access to local measurements. Nevertheless, they remain costly and face important scaling limitations, especially as aircraft components increase in size. Reproducing full-scale icing conditions often requires preserving several competing similarity parameters, including the relationships between droplet size, leading-edge radius, liquid water content, velocity, temperature, and accretion time. In practice, the model geometry or operating conditions must often be adjusted to fit within the facility constraints, and these adjustments can themselves require dedicated extensive analyses [20].

In addition, ice accretion experiments are subject to variability from one test to another, owing to uncertainties in cloud calibration, droplet distributions, thermal conditions, surface roughness, and measurement procedures. These limitations have motivated the development of numerical icing tools as a complementary approach to flight and wind-tunnel testing.

1.4 Ice Accretion Numerical Modeling

Ice accretion is inherently multiphysics, involving the coupling of several physical processes, including the aerodynamic flow around the aircraft, the transport and impingement of super-cooled water droplets, the local thermodynamic balance at the surface, and the progressive modification of the geometry due to ice growth.

Although ice accretion is an unsteady phenomenon, most computational icing solvers rely on a quasi-steady approximation. This assumption is justified by the separation between the short characteristic time scales of the airflow and droplet motion and the much longer time scale associated with ice growth. In practice, the total exposure time is divided into one or several discrete accretion steps. For each step, the flow solution is first computed on the current geometry, followed by the droplet impingement field, the thermodynamic ice-growth rate, and finally the update of the iced surface, as illustrated in Figure 1.3.

The simplest implementation consists of performing the entire accretion process in a single step. This approach is attractive because it limits the computational cost and avoids repeated geometry and mesh updates. However, it assumes that the flow, droplet trajectories, and surface thermodynamic solutions remain representative throughout the full exposure time, which can reduce the accuracy of the predicted ice shape. A more representative strategy consists of using a multilayer approach, in which the geometry is updated several times

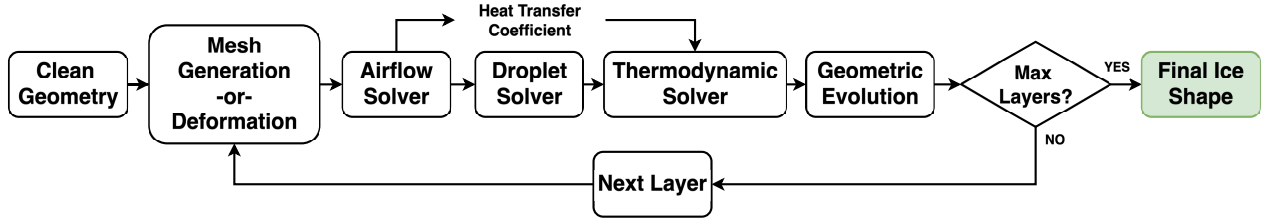


Figure 1.3 Schematic of the multilayer ice accretion framework.

during the accretion process and the physical fields are recomputed on the evolving surface. This improves the coupling between ice growth and the surrounding flow, but also increases the computational cost and introduces additional robustness issues. In particular, repeated geometry deformation or remeshing may progressively deteriorate the mesh quality, leading to highly skewed, tangled, or invalid elements that can interrupt the simulation.

1.5 Problem Statement

Two-dimensional icing simulations have played a central role in the development and validation of ice accretion models. They remain computationally efficient and are particularly useful for airfoil configurations, parametric studies, and comparisons with well-controlled experimental databases. In some applications, they are also extended to approximate three-dimensional effects through 2.5D approaches [21], for example by applying sectional corrections on swept wings. However, such methods remain limited when the ice accretion process is governed by genuinely three-dimensional flow features, spanwise transport, complex geometry effects, or non-uniform impingement patterns.

Fully three-dimensional simulations provide a more general framework for predicting ice accretion on realistic aircraft components and complex configurations. Nevertheless, they are significantly more demanding in terms of computational resources and numerical robustness. These difficulties become particularly important in multilayer simulations, where the geometry must be updated repeatedly as the ice grows. Each update may degrade the mesh quality or generate invalid surface and volume elements. As a result, a simulation that has already required substantial computational effort may fail before reaching the final accretion time.

This robustness constraint partly explains why single-step ice accretion strategies remain widely used in practical applications. They are simpler to execute and reduce the risk of mesh failure, but they also limit the ability of the solver to account for the feedback between the evolving ice shape and the surrounding flow and droplet fields. In the Ice Prediction Workshop

(IPW)³, many submitted ice-shape predictions relied on such single-step approaches. While effective from a computational standpoint, these methods generally showed reduced ability to reproduce complex accretion features, especially when compared with multilayer simulations. This highlights the need for numerical strategies capable of preserving the robustness of single-step approaches while retaining the improved physical fidelity of multilayer ice-growth modeling.

1.5.1 Motivation for the Proposed Approach

The preceding discussion identifies the repeated update of the iced geometry as a major limitation of three-dimensional multilayer ice accretion simulations. Although the flow, droplet, and thermodynamic solutions must be recomputed consistently as the ice shape evolves, the ability to reach the final accretion time often depends on the robustness of the geometry-update procedure itself.

This motivates the use of geometry representations that are less sensitive to local surface irregularities. Previous studies [24–26] have demonstrated the potential of level-set descriptions for representing and evolving iced geometries in two-dimensional configurations. In a level-set formulation, the interface is not tracked explicitly by the surface mesh. Instead, it is represented implicitly as the zero level of a scalar function ϕ , as illustrated in Figure 1.4. The sign of ϕ distinguishes the ice and air regions, while the interface is recovered from the condition $\phi = 0$.

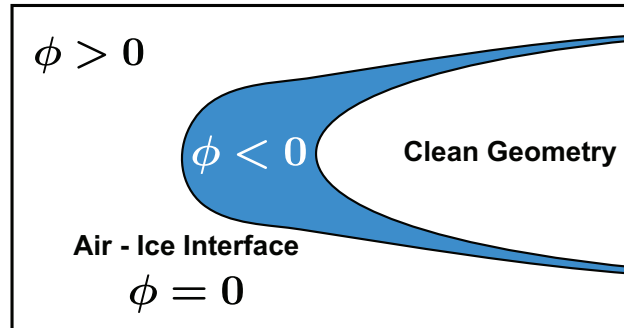


Figure 1.4 Representation of the level-set function ϕ . The ice region is defined by $\phi < 0$, the air region by $\phi > 0$, and the interface by the zero level set $\phi = 0$.

Building on these first attempts, the present work extends this idea to three-dimensional multilayer ice accretion simulations, where geometry updates become more challenging due

³The Ice Prediction Workshops (IPW) [22, 23] is an AIAA international benchmark initiative for evaluating numerical aircraft icing tools against experimental data.

to spanwise variations and increasingly complex ice features. By representing the iced surface implicitly, the geometry can be displaced and reconstructed in a more systematic manner, while preserving the clean portions of the geometry that are not modified by ice accretion. This reconstruction strategy reduces the sensitivity of the workflow to local geometric defects and provides greater control over the quality of the reconstructed surface mesh.

Since geometry evolution is only one component of the icing simulation chain, the preceding physical models must also provide reliable inputs to the accretion process. In particular, droplet impingement predictions determine the local mass flux supplied to the thermodynamic model and directly influence the ice-growth rate. For this reason, Eulerian and Lagrangian droplet formulations, including their extension to Supercooled Large Droplet conditions, are also examined in this thesis.

Finally, immersed-boundary methods are considered in order to reduce the dependence on body-fitted volume remeshing. In a body-fitted approach, the computational mesh conforms to the solid surface, as shown in Figure 1.5a. In contrast, an immersed-boundary method embeds the geometry within a non-conformal background mesh, as shown in Figure 1.5b. The geometry is therefore represented independently of the mesh, which can simplify mesh generation and geometry handling for complex or evolving iced surfaces. The approach is assessed by comparing its aerodynamic, droplet-impingement, and ice-accretion predictions with those obtained using established body-fitted formulations.

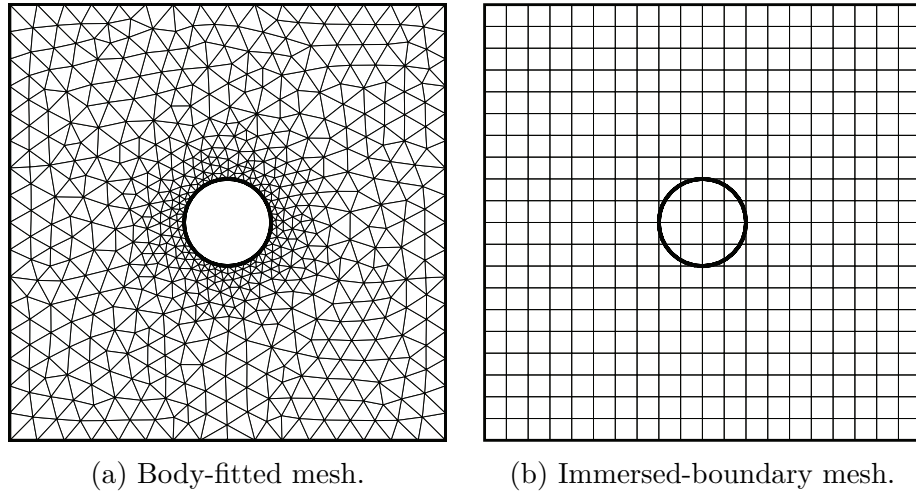


Figure 1.5 Mesh representation around a cylinder using (a) a body-fitted mesh and (b) an immersed-boundary mesh. In the body-fitted approach, the mesh conforms to the cylinder surface, whereas in the immersed-boundary approach, the cylinder is embedded within a non-conformal background mesh.

1.6 Objectives of the Present Study

The primary objective of this thesis is to improve the robustness of three-dimensional multilayer ice accretion simulations through level-set-based geometry evolution. The proposed developments aim to reduce geometric failures during the accretion process, improve the quality of reconstructed iced surfaces, and maintain reliable aerodynamic, droplet-impingement, thermodynamic, and ice-shape predictions throughout the icing simulation chain.

More specifically, this work develops and assesses numerical methods required to support robust multilayer ice accretion simulations, including Lagrangian particle tracking for droplet impingement prediction, Supercooled Large Droplet effects in Eulerian and Lagrangian formulations, level-set-based geometry update and surface reconstruction, and an immersed-boundary extension of the existing icing framework.

To accomplish this objective, the following sub-objectives are defined.

Droplet Impingement Modeling and Supercooled Large Droplet Effects

O.1 Develop, accelerate, and assess Lagrangian particle-tracking methods for droplet impingement prediction

The first objective is to develop and evaluate Lagrangian particle-tracking methods for predicting droplet trajectories and collection efficiency. The method is assessed in terms of impingement limits, collection-efficiency distributions, and computational cost, and is compared with the existing Eulerian droplet model.

O.2 Incorporate Supercooled Large Droplet effects in Eulerian and Lagrangian formulations

The second objective is to account for Supercooled Large Droplet effects in both Eulerian and Lagrangian impingement models. These developments are used to assess the influence of large-droplet effects on droplet collection, impingement limits, and ice accretion.

Multilayer Ice-Shape Evolution

O.3 Improve the robustness of three-dimensional multilayer ice accretion simulations using a level-set formulation

The third objective is to improve the robustness of multilayer ice-shape evolution using a level-set description of the evolving geometry. The method is intended to support geometry displacement, surface reconstruction, and preservation of the clean portions of the geometry,

while reducing geometric failures during multilayer simulations.

O.4 Evaluate the level-set-based methodology in representative icing configurations

The fourth objective is to apply the level-set-based multilayer approach to representative two-dimensional and three-dimensional icing configurations. These cases are used to evaluate the accuracy, robustness, computational efficiency, and reconstructed surface mesh quality of the proposed geometry-evolution strategy.

Immersed-Boundary Formulation and Solver Integration

O.5 Explore immersed-boundary formulations for ice accretion simulations on evolving geometries

The fifth objective is to integrate immersed-boundary formulations for aerodynamic flow, droplet impingement, and ice accretion calculations within the existing numerical framework. This approach is explored in order to reduce the dependence on body-fitted volume remeshing and to simplify mesh generation and geometry handling for complex or evolving iced surfaces.

O.6 Assess the immersed-boundary approach against body-fitted formulations

The sixth objective is to assess the ability of the immersed-boundary approach to reproduce body-fitted predictions for selected verification and validation cases. The comparison focuses on droplet collection efficiencies, and final ice shapes in order to identify the accuracy and limitations of the proposed formulation.

1.7 Thesis Outline

This thesis is organized as a thesis by articles. Chapter 2 first presents the state of the art in numerical ice accretion modeling. It reviews the main components of icing simulation workflows, including aerodynamic modeling, droplet impingement, thermodynamic ice-growth models, SLD effects, geometry evolution strategies, immersed-boundary methods, and level-set techniques.

The first article focuses on Lagrangian particle tracking for droplet impingement prediction under Supercooled Large Droplet conditions. Particular attention is given to the treatment of large-droplet effects and to the ability of the method to predict impingement limits and collection-efficiency distributions.

The second article addresses the problem of robust geometry evolution in three-dimensional multilayer ice accretion simulations. A level-set-based approach is developed and assessed

as a means of updating complex ice shapes while reducing the risk of mesh degradation and invalid geometries during the accretion process.

The third article combines the developments of the previous chapters within an immersed-boundary framework for ice accretion simulations. In this formulation, the aerodynamic and droplet calculations are performed on a background mesh, while the evolving iced geometry is handled through level-set-based reconstruction and update procedures. The objective is to evaluate whether such a framework can reduce the meshing constraints traditionally associated with body-fitted multilayer icing simulations.

Finally, the thesis concludes with a synthesis of the main contributions, followed by a discussion of the limitations of the proposed methods and recommendations for future work.

CHAPTER 2 LITERATURE REVIEW

This chapter reviews the main numerical ingredients involved in aircraft ice accretion simulations. The first section focuses on water-droplet tracking techniques, with particular attention to Eulerian and Lagrangian formulations used to predict droplet impingement. The second section discusses Supercooled Large Droplet (SLD) effects and the semi-empirical models commonly used to account for splashing, bouncing, and secondary droplet behavior. The final sections review immersed-boundary methods, surface-representation techniques, and wall-function approaches relevant to robust ice accretion simulations.

2.1 Water-Droplet Tracking

Water-droplet tracking is a central component of aircraft icing simulations. Before ice growth can be computed, the locations and rates at which droplets impinge on the aircraft surface must be determined. This impingement distribution directly affects the local water mass flux, the surface thermodynamic balance, and ultimately the predicted ice shape. Over the past decades, several numerical methods have been developed to model droplet transport in icing conditions. These methods generally seek a balance between accuracy, computational cost, robustness, and ease of integration into existing flow solvers.

Two main families of approaches are commonly used in the literature: Eulerian and Lagrangian methods. Eulerian models treat the droplet phase as a continuous medium. Governing equations similar to those used for the airflow are solved to obtain the droplet velocity field and the local water concentration throughout the computational domain. This formulation offers several advantages. It is generally computationally efficient, naturally suited for parallelization, and relatively easy to integrate into a finite-volume flow-solver framework. However, Eulerian methods also present limitations. Since the droplet phase is represented as a continuum, individual droplet information is lost, and phenomena such as splashing, breakup, and secondary droplet motion are more difficult to model directly. In addition, artificial dissipation and numerical oscillations can affect the solution, sometimes leading to nonphysical values such as negative water concentration. Such issues must be treated carefully, especially for complex geometries such as multi-element airfoils. Several techniques have been proposed to improve the robustness and conservation properties of Eulerian droplet solvers [27].

Lagrangian methods, on the other hand, track droplets or droplet parcels individually by

solving their equations of motion along their trajectories. Historically, these methods were limited by their computational cost, especially when a large number of droplets was required to obtain smooth impingement maps. However, advances in high-performance computing have renewed interest in Lagrangian approaches for aircraft icing applications. These methods naturally provide access to individual droplet trajectories, impact velocities, and impact angles, which makes them particularly attractive for studying SLD effects and droplet-wall interactions.

The choice between Eulerian and Lagrangian droplet tracking therefore depends on the targeted application and on the acceptable trade-off between computational efficiency, accuracy, and physical detail.

2.1.1 Eulerian Droplet Tracking

Eulerian dispersed-phase tracking is based on a system of partial differential equations describing the evolution of the droplet concentration and droplet momentum. This formulation was first introduced for aircraft icing applications by Bourgault et al. [28]. The continuity equation is solved to obtain the nondimensional water concentration α :

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \mathbf{u}_d) = 0, \quad (2.1)$$

where $\mathbf{u}_d$ is the droplet velocity. The corresponding momentum equation provides the droplet velocity field:

$$\frac{\partial(\alpha \mathbf{u}_d)}{\partial t} + \nabla \cdot (\alpha \mathbf{u}_d \otimes \mathbf{u}_d) = \frac{C_D, Re_D}{24K} \alpha (\mathbf{u}_a - \mathbf{u}_d) + \alpha \left(1 - \frac{\rho_a}{\rho_w} \right) \frac{\mathbf{g}}{Fr^2}. \quad (2.2)$$

In this model, only drag and gravity forces are considered. The first term on the right-hand side of Equation 2.2 represents the aerodynamic drag exerted by the airflow of velocity $\mathbf{u}_a$ on the droplet phase. The second term accounts for gravity.

Eulerian droplet tracking offers several advantages. It is computationally efficient because individual droplets are not tracked separately. It is also well suited for high-performance computing, since the numerical structure of the droplet equations is similar to that of the airflow equations. This similarity facilitates implementation in existing finite-volume solvers and allows the same parallelization strategy to be reused.

However, the Eulerian formulation also has important limitations. Since the droplet phase is averaged over the computational cells, individual droplet trajectories are not directly avail-

able. Microscale phenomena, such as droplet splashing, bouncing, and breakup, are difficult to represent without additional models. Numerical diffusion may also smear the impingement limits and reduce the accuracy of the local collection efficiency. Finally, for low droplet volume fractions or complex configurations, the dispersed phase may be poorly resolved, leading to nonphysical oscillations or negative values of the water concentration.

2.1.2 Lagrangian Droplet Tracking

In a Lagrangian formulation, droplets or droplet parcels are tracked by integrating their equation of motion. The droplet velocity $\mathbf{u}_d$ is first computed, and the droplet position is then obtained through an additional integration step. The equation of motion can be written as

$$\frac{d\mathbf{u}_d}{dt} = \frac{\mathbf{f}}{m} + \mathbf{g} = F(\mathbf{u}_a - \mathbf{u}_d) + \mathbf{g}, \quad (2.3)$$

where $\mathbf{f}$ is the aerodynamic force acting on the droplet, m is the droplet mass, $\mathbf{g}$ is the gravitational acceleration, and F is a scalar coefficient associated with drag.

The main assumptions used in the present Lagrangian formulation are as follows:

- The droplet mass remains constant before impact;
- Droplets are independent from one another;
- The droplet concentration is sufficiently low that the dispersed phase does not affect the airflow;
- Turbulence effects on the droplet trajectories are neglected;
- Droplet rotation and lift forces are neglected.

Under these assumptions, the only forces considered are aerodynamic drag and gravity. The scalar coefficient F is obtained from the local droplet Reynolds number, based on the relative velocity $|\mathbf{u}_a - \mathbf{u}_d|$ and droplet diameter. This gives

$$F = \frac{C_D \frac{1}{2} \rho_a |\mathbf{u}_a - \mathbf{u}_d|^2 A}{m |\mathbf{u}_a - \mathbf{u}_d|} = \frac{3C_D \rho_a |\mathbf{u}_a - \mathbf{u}_d|}{4\rho_w D}, \quad (2.4)$$

where C_D is the droplet drag coefficient, ρ_a is the air density, ρ_w is the water density, and D is the droplet diameter.

A detailed analysis of numerical schemes for ordinary differential equations is presented by Darmofal and Haimes [29]. Multistep schemes, such as Adams-Bashforth methods, and multistage schemes, such as Runge-Kutta methods, can be used to integrate droplet trajectories. However, particle-path integration may become unstable if the time step is too large. Therefore, a compromise must be found between computational cost, accuracy, and stability.

In aircraft icing applications, the trajectory integration also depends strongly on interpolation of the airflow velocity. Since droplet paths do not generally coincide with the computational mesh, the host cell of each droplet must be identified after each integration step. Several localization and tracking techniques have been proposed for this purpose.

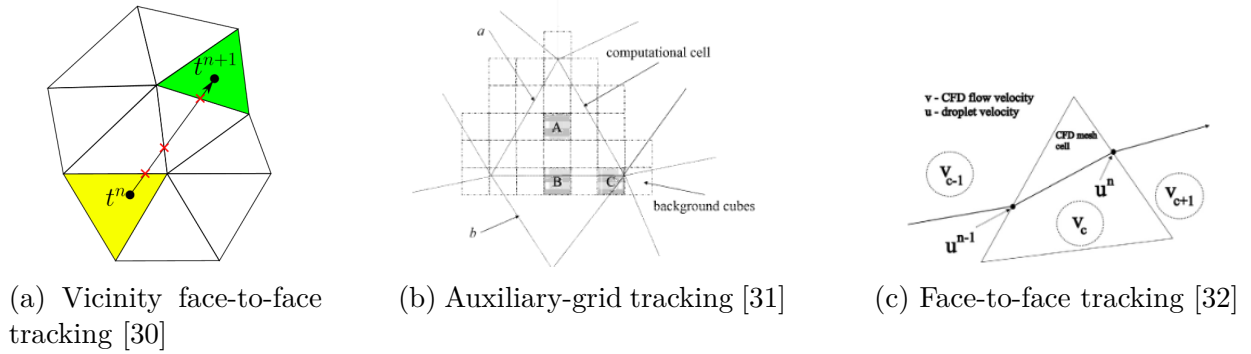


Figure 2.1 Examples of droplet-tracking techniques.

Tree-based search methods use octree-type data structures to identify the cell containing each droplet. Although general, these methods can become expensive for large-scale three-dimensional simulations. For example, a swept-wing dispersed-phase simulation may require several million droplets to obtain a sufficiently resolved impingement distribution. In this context, computational cost and parallel scalability are critical, which limits the practical use of global tree-based searches.

Known-vicinity face-to-face methods [30] use the previous droplet location to identify the current host cell. After the droplet position is advanced, a ray is cast along the droplet trajectory. If the trajectory remains within the current cell, the host cell is unchanged. Otherwise, the droplet crosses a face into an adjacent cell, which is identified using mesh connectivity. This procedure is repeated until the final host cell is found. This approach avoids global searches and is therefore more efficient for large unstructured grids.

Auxiliary-grid methods [31] use a structured background grid, typically Cartesian, to accelerate droplet localization. Before the simulation, a mapping is established between the background grid and the computational mesh. Each background cell stores the computa-

tional cells that it overlaps. During tracking, the droplet is first located in the auxiliary grid, and point-in-cell tests are then performed only on a small set of candidate computational cells. This can significantly reduce the cost of particle localization. However, the construction and storage of the background-grid mapping introduce additional preprocessing complexity, especially in three-dimensional simulations.

Face-to-face tracking [32] provides an attractive compromise between efficiency and accuracy. The method relies directly on mesh connectivity. At each step, the droplet velocity defines a trajectory segment, and the next cell is identified by finding the nearest face intersection. The local time step is then obtained from the residence time of the droplet in the current cell. This approach avoids explicit global cell searches and is naturally compatible with finite-volume data structures. It is therefore well suited for large-scale unstructured simulations.

Because stability is a significant concern in droplet trajectory integration, implicit schemes are often preferred. Backward differentiation formulas provide an efficient and robust option [32]. Further details on the implementation used in the present work are provided in Chapter ??.

Using a first-order backward difference formula, the discretized equation of motion can be written as

$$\alpha m \mathbf{u}_d^n + \beta m \mathbf{u}_d^{n-1} = \mathbf{f}(\mathbf{u}_a, \mathbf{u}_d) + m \mathbf{g} = m F(\mathbf{u}_a - \mathbf{u}_d) + m \mathbf{g}, \quad (2.5)$$

where $\alpha = 1/\Delta t$ and $\beta = -1/\Delta t$. The droplet velocity at the new time level is then obtained from

$$\mathbf{u}_d^n = \frac{F \mathbf{u}_a - \beta \mathbf{u}_d^{n-1} + \mathbf{g}}{\alpha + F}. \quad (2.6)$$

A second-order formulation can be obtained by retaining the velocity from the previous two time levels:

$$\mathbf{u}_d^n = \frac{F \mathbf{u}_a - \beta \mathbf{u}_d^{n-1} - \gamma \mathbf{u}_d^{n-2} + \mathbf{g}}{\alpha + F}. \quad (2.7)$$

For variable time steps, the coefficients α , β , and γ are computed using the current time step Δt_c and the previous time step Δt_{c-1} :

$$\alpha = \frac{2\Delta t_c + \Delta t_{c-1}}{(\Delta t_c + \Delta t_{c-1})\Delta t_c}, \quad (2.8)$$

$$\beta = \frac{-(\Delta t_c + \Delta t_{c-1})}{\Delta t_c \Delta t_{c-1}}, \quad (2.9)$$

$$\gamma = \frac{\Delta t_c}{(\Delta t_c + \Delta t_{c-1}) \Delta t_{c-1}}. \quad (2.10)$$

The trajectory integration is performed iteratively. An initial estimate of the new droplet velocity is obtained from the previous value, $\mathbf{u}_d^n = \mathbf{u}_d^{n-1}$. The airflow velocity at the new position is then estimated, and the nearest face intersection is computed using the current trajectory estimate. This provides the local time step and the corresponding coefficients in Equations 2.8–2.10. Equation 2.6 or Equation 2.7 is then used to update the droplet velocity. The process is repeated until the change between successive velocity estimates becomes sufficiently small.

The face intersection is computed using the midpoint velocity

$$\mathbf{u}_d^{n-\frac{1}{2}} = \frac{1}{2} (\mathbf{u}_d^n + \mathbf{u}_d^{n-1}). \quad (2.11)$$

Because the local time step may vary significantly from one cell to another, the method may locally reduce to first-order accuracy. To mitigate this issue, limited gradients can be used to interpolate the airflow velocity at the droplet position:

$$\mathbf{u}_{a\text{facet}} = \mathbf{u}_{a\text{cell}} + \nabla \mathbf{u}_a \cdot \mathbf{d}, \quad (2.12)$$

where $\mathbf{d}$ is the vector from the cell center to the droplet position.

The limiting behavior of Equation 2.4 provides useful validation cases. For an infinitesimally small droplet, the coefficient F tends to infinity, and the droplet velocity approaches the airflow velocity, $\mathbf{u}_d^n \approx \mathbf{u}_a^n$. In this limit, droplet trajectories coincide with airflow streamlines. Conversely, for very large droplets, F tends toward zero, and the airflow has little influence on the droplet motion. The trajectory is then mainly governed by inertia and gravity.

2.1.3 Collection Efficiency

In addition to impingement locations, droplet tracking provides the collection efficiency of the geometry. The collection efficiency β is a nondimensional measure of the local impingement rate. It relates the amount of water captured by a surface region to the amount of water that would pass through the corresponding upstream area, as illustrated in Figure 2.2:

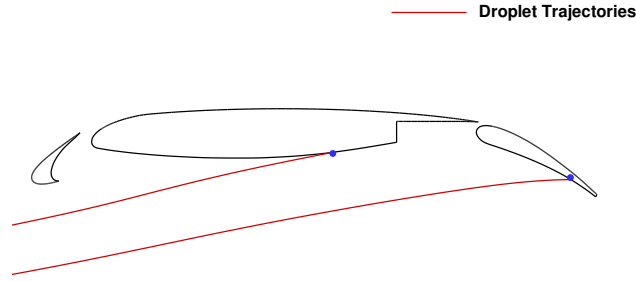


Figure 2.3 Pathological case for collection-efficiency computation on a multi-element airfoil.

To address these issues, Xie et al. [2] proposed computing additional secondary trajectories around each primary droplet. Since the secondary trajectories are generated locally, they can be stored on the same processor, reducing communication requirements. The droplets are also released very close to one another, which increases the probability that they impinge on the same component of the geometry.

2.2 Supercooled Large Droplet Modeling

Supercooled Large Droplet conditions have received significant attention following the introduction of Appendix O to the Federal Airworthiness Regulations. Compared with standard icing conditions, SLD environments involve larger droplets with greater inertia. These droplets are less affected by the airflow and are more likely to impinge farther downstream on the aircraft surface. In addition, SLD impacts may involve splashing, bouncing, breakup, and secondary droplet reimpingement. These phenomena can extend the impingement limits and generate ice accretion in regions where ice protection systems may not be installed.

SLD ice accretion is challenging from both experimental and numerical perspectives. Experimentally, obtaining reliable data is difficult, time-consuming, and expensive. Papadakis et al. [33–37] provided extensive data on SLD impingement distributions and ice shapes. Other studies focused on specific SLD mechanisms, such as droplet deformation, breakup, and impact behavior [38–40]. Broeren et al. [3] identified several characteristic features of SLD ice accretion, including horns, nodules, and clear ice, as shown in Figure 2.4.

SLD behavior differs significantly from that of supercooled standard droplets. Because of their larger diameter, SLDs have greater inertia and their trajectories deviate more strongly from the airflow streamlines. Wright et al. [41] classified SLD effects according to their

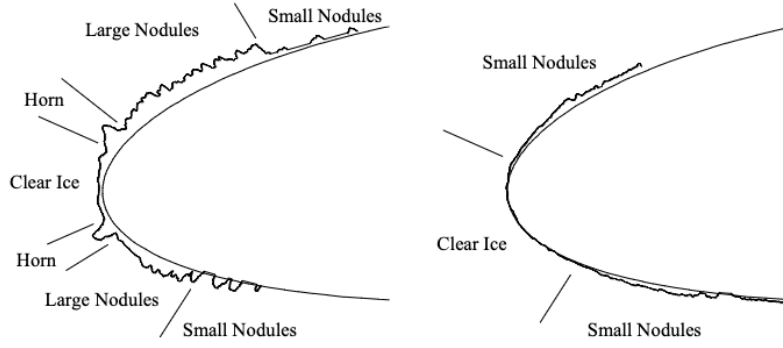


Figure 2.4 Main features of SLD ice accretion [3].

relative importance for aircraft icing simulations.

2.2.1 Third-Order Effects

Third-order SLD effects are generally small and are often neglected in practical icing simulations. This assumption is consistent with the hypotheses adopted in Section 2.1.2. These effects include the following:

- **Basset force:** This force accounts for the time history of the drag force. It is generally negligible when the viscosity of the carrier fluid is much smaller than that of the particles, as is the case for water droplets in air;
- **Saffman force:** This lift force can be relevant for small particles in shear flows, but is typically more important for gas bubbles than for water droplets in icing applications;
- **Turbulence effects:** Bhargava et al. [42] investigated turbulence effects on droplet motion in the NASA IRT and found that they are generally of secondary importance, except in specific high-speed configurations such as rotor blades or propellers.

2.2.2 Second-Order Effects: Deformation and Breakup

Unlike standard droplets, SLDs may deform significantly as they travel through the flow field. Because of their larger diameter, they are more sensitive to aerodynamic pressure differences across the droplet surface. When the aerodynamic forces exceed the restoring effects of surface tension and viscosity, the assumption of spherical droplets becomes invalid.

Droplet deformation changes the effective drag coefficient and can therefore influence the droplet trajectory and the resulting impingement distribution. In more extreme cases,

droplets may break up into smaller fragments, further modifying the droplet-size distribution and the water collection on the surface. However, except for some freezing-drizzle conditions, the influence of deformation and breakup appears to remain limited in many icing applications [43].

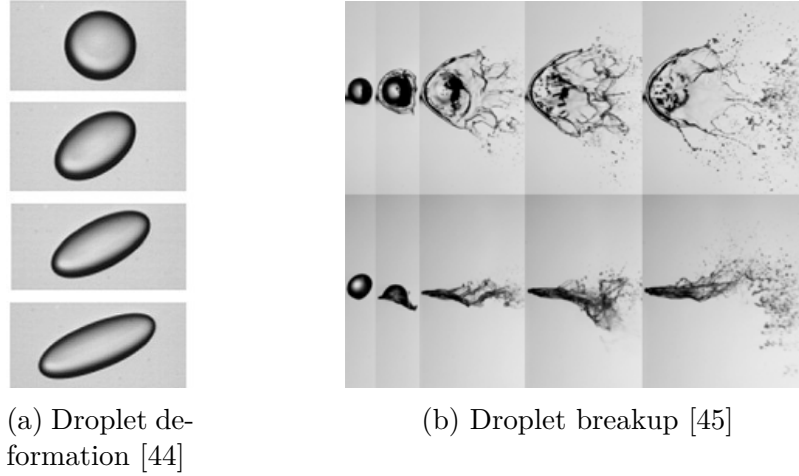


Figure 2.5 Examples of second-order SLD effects.

2.2.3 First-Order Effects: Sticking, Splashing, and Bouncing

Droplet-wall interaction is one of the most important aspects of SLD modeling. When large droplets impact a surface, they do not necessarily stick completely. Depending on the impact conditions and surface properties, they may spread, bounce, splash, or generate secondary droplets. Experiments have shown that large droplets may fragment after impact, producing secondary droplets that can reimpinge farther downstream [46–49]. This mechanism increases the effective impingement limits compared with standard droplets.

The outcome of droplet impact depends on several parameters, including droplet diameter, impact velocity, impact angle, density, viscosity, surface tension, surface roughness, wettability, and wall temperature. These quantities are commonly expressed using nondimensional groups such as the Reynolds, Weber, and Ohnesorge numbers. The main droplet-wall interaction regimes can be summarized as follows [50]:

- **Sticking:** The droplet adheres to the wall, and no mass loss occurs. This regime is generally associated with low impact velocity;
- **Bouncing:** A thin air film remains trapped between the droplet and the surface, preventing direct contact and causing the droplet to rebound;

- **Deposition and spreading:** The droplet spreads along the surface, contributing to a liquid film or merging with an existing water film. Part of the kinetic energy is dissipated by viscous effects, while another part is converted into surface energy;
- **Splashing:** At sufficiently high impact energy, viscous forces and surface tension are no longer able to maintain droplet integrity. The droplet fragments into secondary droplets, some of which are ejected from the surface.

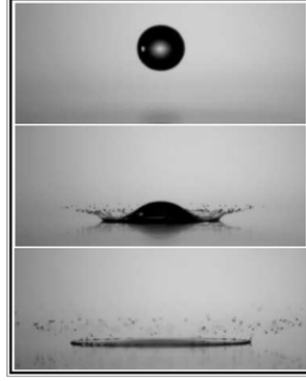


Figure 2.6 Example of droplet splashing [4].

Direct numerical simulation of droplet impact can be performed using interface-resolved methods such as the Volume of Fluid method [51] or Moment of Fluid methods [52]. However, such approaches require high-resolution meshes around each droplet and are therefore not practical for aircraft icing simulations involving millions of droplets. As a result, current SLD models generally rely on semi-empirical formulations calibrated against experimental data.

These models often introduce a mass-loss coefficient f_m , or equivalently a sticking efficiency ϵ , to account for the fraction of droplet mass retained after impact. The corrected collection efficiency is then written as

$$\beta_{final} = (1.0 - f_m), \beta_0, \quad (2.15)$$

or equivalently,

$$\beta_{final} = \epsilon, \beta_0. \quad (2.16)$$

This correction can be applied as a post-processing step for both Eulerian and Lagrangian droplet models. However, secondary droplet reimpingement is more naturally handled in a

Lagrangian framework. In this case, semi-empirical models can provide post-impact quantities, such as secondary droplet diameter and velocity, allowing the droplets to be reinjected into the computational domain.

In an Eulerian framework, secondary reimpingement is more difficult to model. One possible approach consists in modifying the boundary conditions on splashing surfaces [53]. The simulation is then performed in two stages. First, the original collection efficiency is computed and the splashing regions are identified. Second, selected wall faces are converted into inflow boundaries to model the reemission of secondary droplets. The final collection efficiency is then obtained as

$$\beta_{final} = \epsilon, \beta_0 + \sum \beta_{re-impingement}. \quad (2.17)$$

For complex three-dimensional geometries, however, the number of surface elements involved in reinjection may become very large, reducing the practical efficiency of this approach. Recent work has proposed converting all relevant boundary faces into inflow boundaries to reduce the computational cost [54]. Other approaches use statistical clustering to reduce the number of reinjection simulations [53].

Table 2.1 summarizes several splashing models available in the literature. Mundo et al. [55] proposed one of the first semi-empirical models for droplet splashing, in which the mass-loss coefficient is related to the number and size distribution of secondary droplets. Trujillo et al. [56] introduced a model based on the parameters proposed by Yarin and Weiss [57], including effects of droplet-generator frequency and surface roughness. Honsek et al. [58] later modified Trujillo’s formulation and implemented it in FENSAP-ICE.

Wright et al. [41] introduced the impact angle into the splashing parameters and mass-loss formulation. This makes splashing less likely for droplets impacting nearly normal to the surface and more likely near the impingement limits. Their model was calibrated by comparing computed water collection efficiencies with experimental data. Trontin et al. [43] proposed an alternative formulation based on subfunctions describing the dependence of mass loss on impact angle and kinetic energy. The model parameters were adjusted using comparisons with experimental data.

2.3 Immersed-Boundary Methods

The robustness of aircraft icing simulations is strongly affected by the geometry and mesh-update stages. In classical body-fitted approaches, the computational mesh must conform to

Table 2.1 SLD Splashing Models

Source	Mass Loss Coefficient	Splashing Criterion
Mundo et al. [55]	$n_s(\frac{d_s}{d_0})^3$	$K_M = (Oh^{-0.4}We)^{0.625} > 57.7$
Trujillo et al. [56]	$0.8[1.0 - e^{-0.85(K_y-17)}]$	$K_y = \frac{1}{0.45^{\frac{3}{8}}}(Oh^{-0.4}We)^{0.3125} > 17$
Honsek et al. [58]	$\frac{3.8}{K_y^{0.5}}[1.0 - e^{-0.85(K_y-17)}]$	$K_y > 17$
Wright et al. [41]	$0.7(1.0 - \sin(\theta_0))[1.0 - e^{-0.0092(K_M-200.0)}]$	$\frac{0.859 K_m^{0.5} (\frac{\rho_w}{LWC})^{0.125}}{\sin(\theta_0)^{1.25}} > 200.0 \ \& \ \theta < \frac{\pi}{2}$
Trontin et al. [43]	$1.0 - g(\frac{\theta}{\theta_c})f(\frac{K_c-K_0}{K_0})$	$K_c = We_n \cdot Oh^{\frac{-2}{5}} > 657$

the aircraft surface and must be regenerated or deformed after each ice accretion layer. As the ice shape grows, the geometry can become increasingly complex, making it difficult to maintain high-quality near-wall cells and avoid mesh failure. This issue is especially critical for three-dimensional multilayer simulations.

Immersed-boundary methods provide an alternative approach in which the geometry is represented independently of the computational mesh. The solid boundary may cut through grid cells, and boundary conditions are imposed through modifications of the governing equations or their discretization. This strategy can simplify mesh generation and improve robustness for complex and evolving geometries. However, each immersed-boundary formulation introduces specific numerical challenges related to accuracy, conservation, stability, wall modeling, and geometry representation.

2.3.1 Classification

Following the classification of Sotiropoulos and Yang [59], immersed-boundary methods can be grouped into three main families: geometrical methods, continuous-forcing methods, and discrete-forcing methods. Geometrical methods modify the mesh or control volumes near the interface. Continuous-forcing methods add source terms to the governing equations to represent the boundary. Discrete-forcing methods impose the boundary conditions directly

at the discrete level. The following sections summarize these approaches. Table 2.2 provides a comparison based on implementation complexity, accuracy, stability, and computational cost.

Geometrical Methods

Geometrical immersed-boundary methods rely on the ability of the solver to handle arbitrary cell shapes. These methods were introduced by Clarke et al. [5] for Euler-flow simulations around multi-element airfoils. In this approach, a preprocessing step cuts the cells intersected by the solid boundary, as illustrated in Figure 2.7. The resulting cut cells conform locally to the immersed interface.

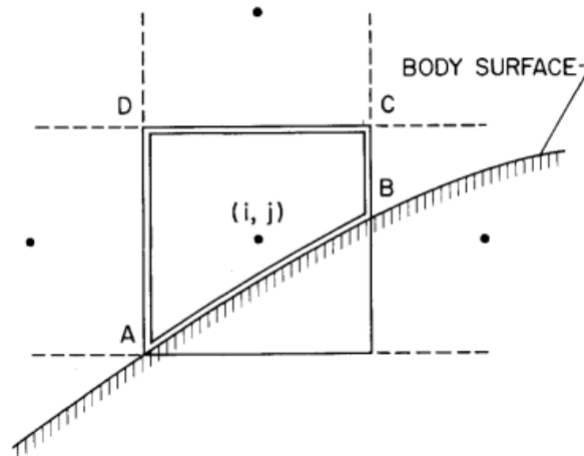


Figure 2.7 Boundary cut-cell method [5].

The main advantage of cut-cell methods is their sharp and conservative treatment of the boundary. In addition, the governing equations and numerical fluxes can often be kept close to those of the original solver. However, the geometric preprocessing is complex, particularly in three dimensions. Another major difficulty is the presence of very small cut cells, which can impose severe time-step restrictions and affect stability. Several remedies have been proposed, including cell merging [60], cell linking [61], and cell mixing [62].

Continuous-Forcing Methods

Continuous-forcing immersed-boundary methods modify the governing equations by adding a forcing term that accounts for the presence of the boundary. A generic system can be written as

$$\frac{\partial \mathbf{W}}{\partial t} = \mathbf{R} + \mathbf{F}, \quad (2.18)$$

where $\mathbf{W}$ represents the conservative variables, $\mathbf{R}$ is the residual of the original governing equations, and $\mathbf{F}$ is the immersed-boundary forcing term.

This class of methods originates from the work of Peskin [63], who introduced an immersed-boundary formulation for elastic boundaries in heart-flow simulations. In this method, the boundary is represented by Lagrangian markers located at positions $\mathbf{X}_k$ and moving with the fluid. The force is distributed to the surrounding grid points using a regularized Dirac function:

$$\mathbf{F} = \sum_k \mathbf{F}_k(t) \delta(\mathbf{r}_k), \quad (2.19)$$

with

$$\mathbf{r}_k = |\mathbf{x} - \mathbf{X}_k|. \quad (2.20)$$

Here, $\mathbf{F}_k(t)$ is the force associated with marker k , typically obtained from a constitutive law. For rigid-body applications, the boundary can be modeled as a very stiff elastic structure. Lai and Peskin [64] used a spring analogy in which the marker is forced toward an equilibrium position $\mathbf{X}_{k,e}$:

$$\mathbf{F}_k(t) = k(\mathbf{X}_k - \mathbf{X}_{k,e}). \quad (2.21)$$

A high stiffness k is required to mimic a rigid boundary, but this also makes the governing equations stiff and can lead to numerical instability. Implicit or semi-implicit treatments are therefore often required. Moreover, because the force is distributed over several grid points, the interface is diffuse, which generally limits the spatial accuracy of the method near the boundary.

Penalization methods provide another continuous-forcing strategy. The solid region is treated as a porous medium with very low permeability. This approach was introduced by Arquis and Caltagirone [65]. A source term is added to the momentum equations to drive the velocity inside the solid toward the solid velocity $\mathbf{u}_s$:

$$\mathbf{F} = \sum_i \lambda \chi_{s,i} (\mathbf{u}_{s,i} - \mathbf{u}), \quad (2.22)$$

where χ is a mask function identifying the solid region and λ is a large penalization parameter. As for stiff elastic forcing, large values of λ increase the stiffness of the equations and may require implicit or semi-implicit time integration.

Discrete-Forcing Methods

Discrete-forcing methods impose the immersed-boundary conditions directly in the discretized equations rather than modifying the continuous governing equations. Direct forcing, introduced by Mohd-Yusof [66], is a representative example. For the incompressible Navier-Stokes equations, the momentum equation can be written as

$$\frac{\partial \mathbf{u}}{\partial t} = -\mathbf{u} \cdot \nabla \mathbf{u} - \nabla P + \frac{1}{Re} \nabla^2 \mathbf{u} + \mathbf{f}, \quad (2.23)$$

where $\mathbf{f}$ is the forcing term. Using an explicit Euler discretization gives

$$\frac{\mathbf{u}^{n+1} - \mathbf{u}^n}{\Delta t} = -\mathbf{u} \cdot \nabla \mathbf{u} - \nabla P + \frac{1}{Re} \nabla^2 \mathbf{u} + \mathbf{f}. \quad (2.24)$$

The forcing term can then be selected to impose a target boundary velocity $\mathbf{u}_{target}$:

$$\mathbf{f} = \mathbf{u} \cdot \nabla \mathbf{u} + \nabla P - \frac{1}{Re} \nabla^2 \mathbf{u} + \frac{1}{\Delta t} (\mathbf{u}_{target} - \mathbf{u}^n). \quad (2.25)$$

If the immersed boundary coincides with a cell center, the target value can be imposed exactly. In practice, this alignment is rare, and interpolation is required to avoid a staircase representation of the boundary. Unlike continuous-forcing methods, discrete-forcing methods do not introduce large source terms into the continuous equations and therefore avoid some stiffness-related stability issues. However, care is required near highly curved boundaries and at high Reynolds numbers, where internal flow structures inside the solid region may affect stability.

Ghost-cell methods are another widely used discrete-forcing approach [6]. In these methods, cells located inside the solid but adjacent to the fluid are identified as ghost cells. For each ghost cell, an image point is constructed in the fluid, usually by casting a ray normal to the interface from the ghost-cell center. Flow variables are interpolated at the image point from neighboring fluid cells, and the ghost-cell values are then assigned so that the desired boundary condition is satisfied at the immersed interface. Figure 2.8 illustrates the construction of ghost cells and image points.

Ghost-cell methods can achieve first- or second-order spatial accuracy, depending on the

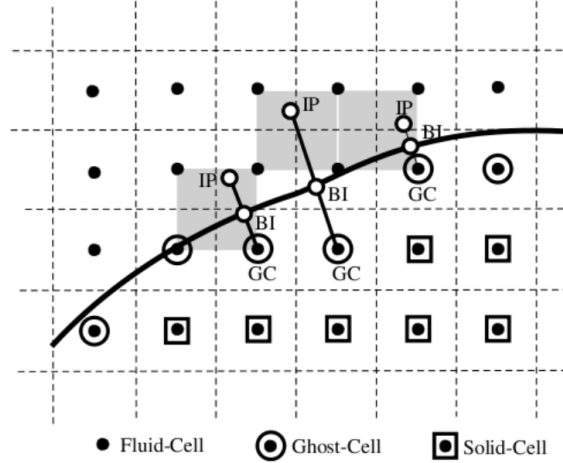


Figure 2.8 Image-point and ghost-cell construction near an immersed boundary [6].

interpolation strategy and the placement of the image point [67]. They also preserve the original form of the governing equations in the fluid region, which makes them attractive for compressible-flow solvers.

Face-forcing methods impose the boundary condition through modified fluxes at cell faces. This approach has been applied to Euler flows [68], RANS simulations [69], and Eulerian droplet tracking [7]. The construction is similar to the ghost-cell approach, but the image point is defined along a normal passing through the cell face rather than the ghost-cell center. The reconstructed state is then used to compute the flux through the immersed boundary face.

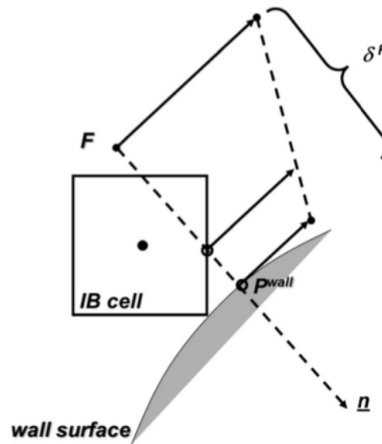


Figure 2.9 Illustration of face forcing near an immersed boundary [7].

Table 2.2 IBM Comparison

Class	Method	Implementation	Accuracy	Stability	Cost
Geometrical	Cut-Cell	Hard	1 st	Hard	High
Continuous Forcing	Rigid Body Forcing	Easy	1 st Order	Hard	Low
	Penalization	Easy	2 nd Order	Hard	Low
	Direct Forcing	Medium	2 nd Order	Hard	Low
Discrete Forcing	Ghost-Cell	Medium	2 nd Order	No Issues	High
	Face Forcing	Medium	2 nd Order	No Issues	High

2.3.2 Surface Representation

Surface representation is an essential component of both ice accretion simulations and immersed-boundary methods. In ice accretion, the surface must be updated as ice grows. In immersed-boundary methods, the interface must be located accurately in order to impose boundary conditions and compute quantities such as normals, wall distances, and curvature.

Two main categories of interface representation exist in the literature: interface tracking and interface capturing. Interface-tracking methods represent the surface explicitly using markers or mesh points that move with the interface. Interface-capturing methods represent the surface implicitly as a field defined over the computational domain.

Interface Tracking

Interface-tracking methods are commonly used for problems involving discontinuous interfaces, such as two-phase flows. The interface is represented by marker points distributed either directly on the interface or within control volumes. Examples include the Marker-and-Cell and Particle-in-Cell methods.

The main advantage of interface tracking is that the interface position is explicitly known, which enables an accurate representation of fine surface structures. However, because the interface markers do not generally coincide with the computational grid, flow quantities must be interpolated from the grid to the markers. This interpolation may introduce errors. In addition, large interface deformations can lead to poor marker distribution, requiring

reseeding or remeshing. Topological changes, such as merging or breakup, are also difficult to handle robustly with explicit interface tracking.

Interface Capturing

Interface-capturing methods represent the interface implicitly using a field variable. One of the most common approaches is the level-set method. In this method, the interface is represented as the zero contour of a scalar function ϕ defined over the computational domain [70]. The level-set function is advected according to

$$\frac{\partial \phi}{\partial t} + \mathbf{V} \cdot \nabla \phi = \phi(\nabla \cdot \mathbf{V}), \quad (2.26)$$

where $\mathbf{V}$ is the interface velocity field. The function ϕ is commonly defined as positive in the fluid, negative in the solid, and zero at the interface. When ϕ is a signed-distance function, interface normals and curvature can be obtained from

$$\mathbf{n} = \frac{\nabla \phi}{|\nabla \phi|}, \quad (2.27)$$

and

$$\kappa = \nabla \cdot \frac{\nabla \phi}{|\nabla \phi|}. \quad (2.28)$$

This information is useful for immersed-boundary methods, geometry evolution, and turbulence modeling.

Level-set methods can naturally handle topological changes and provide smooth interface normals. However, they are not strictly conservative. In addition, during advection, the signed-distance property may be lost, especially when tangential velocity components are present near the interface. A reinitialization or redistancing procedure is therefore often required to restore the signed-distance property. The interface velocity must also be known not only at the interface but in its neighborhood, which may require an additional velocity-extension step.

Another widely used interface-capturing technique is the Volume of Fluid method. In this approach, a volume fraction $f(\mathbf{x}, t)$ is used to represent the fraction of a given material contained in each cell [71]:

$$f(\mathbf{x}, t) = \begin{cases} 0 & \text{fluid,} \\ [0, 1] & \text{interface,} \\ 1 & \text{solid.} \end{cases} \quad (2.29)$$

The interface is reconstructed from the volume fraction in each mixed cell. The accuracy of the method depends strongly on the reconstruction strategy. The Simple Linear Interface Calculation method is first-order accurate, whereas the Piecewise Linear Interface Calculation method can achieve second-order accuracy.

The VoF/PLIC algorithm is generally divided into two steps: reconstruction and propagation. Figure 2.10 illustrates the reconstruction process for SLIC and PLIC. PLIC is usually preferred because it provides a more accurate representation of the interface. However, for highly curved interfaces, features smaller than the local grid size cannot be resolved [8].

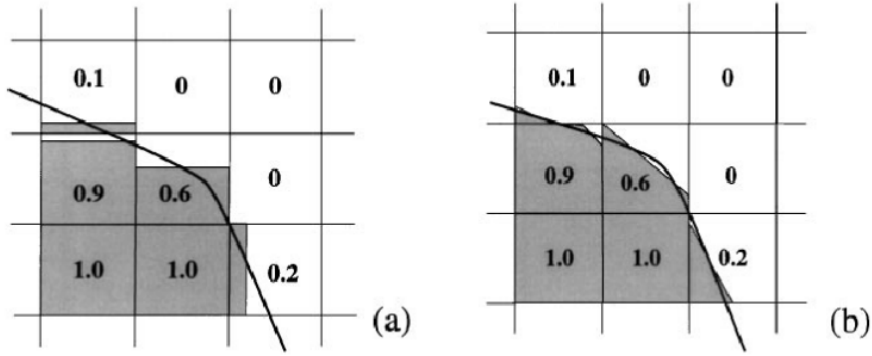


Figure 2.10 Interface reconstruction using (a) SLIC and (b) PLIC [8].

Hybrid methods combining level-set and VoF approaches have also been proposed. The Coupled Level Set Volume-of-Fluid method, introduced by Marter [72], combines the smooth geometric information of the level-set method with the conservation properties of VoF. In this approach, both fields are evolved simultaneously. The VoF field maintains conservation, while the level-set field provides normals, curvature, and distance information. The level-set interface is corrected using the VoF field, and a redistancing step is performed to maintain the signed-distance property.

The main limitation of CLSVOF methods is their implementation complexity and computational cost, since both the level-set and VoF equations must be solved and coupled. Nevertheless, they provide improved capabilities compared with either method used alone.

2.3.3 Wall Functions

At high Reynolds numbers, mean-flow gradients near the wall become very steep. In low-Reynolds-number RANS approaches, the turbulence model is integrated all the way to the wall. This requires very fine near-wall meshes and imposes strong constraints on the first-cell height, orthogonality, and grid smoothness. In ice accretion simulations, these constraints are particularly difficult to maintain because the surface evolves and may develop complex features such as horns, scallops, and runback ice.

Wall functions provide a practical alternative by using semi-analytical near-wall profiles for velocity and temperature. Instead of resolving the entire viscous sublayer, wall functions model the relation between wall quantities and flow variables at a finite distance from the wall. This reduces mesh requirements and can improve the robustness of icing simulations, especially in immersed-boundary or multilayer frameworks.

Near the wall, streamwise gradients are often small compared with wall-normal gradients. For an incompressible steady flow, the RANS momentum equation can be simplified as

$$\frac{d}{dy} \left((\mu + \mu_t) \frac{du}{dy} \right) = \frac{dp}{dx}. \quad (2.30)$$

Assuming a zero pressure gradient and integrating once gives

$$(\mu + \mu_t) \frac{du}{dy} = \tau_w, \quad (2.31)$$

where τ_w is the wall shear stress. In nondimensional form, this becomes

$$(1 + \nu_t^+) \frac{du^+}{dy^+} = 1. \quad (2.32)$$

The inner region of the boundary layer is commonly divided into three subregions:

- **Viscous sublayer** ($y^+ < 5$): viscous effects dominate and turbulent viscosity is negligible. The velocity profile is approximately linear, $u^+ = y^+$;
- **Buffer layer** ($5 \leq y^+ < 30$): viscous and turbulent effects are both important;
- **Logarithmic layer** ($y^+ \geq 30$): turbulent transport dominates. Assuming $\nu_t^+ = \kappa y^+$, where $\kappa \approx 0.41$ is the von Karman constant, the velocity profile follows the logarithmic law.

The standard wall function is therefore written as

$$u^+ = \begin{cases} y^+ & y^+ < 5, \\ \frac{1}{\kappa} \ln(y^+) + C & y^+ \geq 30, \end{cases} \quad (2.33)$$

where C is an empirical constant, typically between 5.0 and 5.5 depending on the turbulence model and calibration.

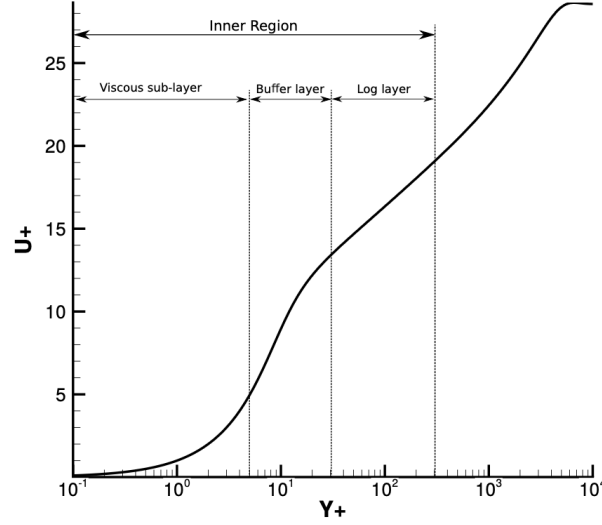


Figure 2.11 Illustration of the different layers of a turbulent boundary layer [9].

A similar analysis can be applied to the energy equation to obtain temperature wall functions. The simplified energy equation can be written as

$$\frac{d}{dy} \left[\left(\frac{\mu}{Pr} + \frac{\mu_t}{Pr_t} \right) \frac{dT}{dy} \right] = 0. \quad (2.34)$$

The corresponding nondimensional temperature profile is

$$T^+ = \begin{cases} Pr, y^+ & y^+ < 5, \\ \frac{Pr_t}{\kappa_T} \ln(y^+) + C_T & y^+ \geq 30, \end{cases} \quad (2.35)$$

where $\kappa_T = 0.48$ and $C_T = 3.9$ are empirical constants [73].

Velocity Wall Functions

Several velocity wall functions have been proposed in the literature. The simplest is the standard wall function given in Equation 2.33. However, the buffer layer is difficult to model analytically because both viscous and turbulent effects are important. This has motivated the development of more general wall laws that interpolate between the viscous sublayer and logarithmic region. Such formulations are often referred to as universal wall functions because they describe the entire inner region. Table 2.3 summarizes several commonly used velocity wall functions.

Table 2.3 Velocity Wall Functions

Source	SA or $k - \omega$ SST	Velocity Profile
Spalding [74]	Both	$y^+ = u^+ + e^{-\kappa B} (1 - e^{\kappa u^+} - \frac{(\kappa u^+)}{1!} - \frac{(\kappa u^+)^2}{2!} - \dots)$
Reichardt [75]	Both	$u^+ = \frac{\ln(1+\kappa y^+)}{\kappa} + 7.8(1 - e^{-\frac{y^+}{11}} - \frac{y^+}{11}e^{-\frac{y^+}{3}})$
Knopp [76]	SA	$F_{SA,\alpha} = (1 - \phi_{SA})F_{Sp,5} + \phi_{SA}F_{Rei,m}$ with $\phi_{SA} = \tanh[(\frac{y^+}{24})^3]$
	$k - \omega$ SST	$F_{k\omega,\alpha} = (1 - \phi_{k\omega})F_{Sp,3} + \phi_{k\omega}F_{Rei,m}$ with $\phi_{k\omega} = \tanh[(\frac{y^+}{50})^2]$
Allmaras [77]	SA	$u^+ = \bar{B} + c_1 \ln((y^+ + a_1)^2 + b_1^2) - c_2 \ln((y^+ + a_2)^2 + b_2^2) - c_3 \operatorname{atan2}(b_1, y^+ + a_1) - c_4 \operatorname{atan2}(b_2, y^+ + a_2)$

More general wall functions have also been developed for nonequilibrium boundary layers. Craft et al. [78] proposed an Analytical Wall Function that accounts for pressure-gradient and convection effects by integrating a more general near-wall momentum equation. This requires an assumed relation between turbulent viscosity and wall-normal distance.

Gant [79] proposed a wall function based on the numerical integration of a parabolized transport equation on a one-dimensional subgrid. The subgrid solution is then interpolated to the first wall cell.

Popovac and Hanjalic [80] observed that many advanced wall functions involve complex expressions that are difficult to implement in general-purpose CFD solvers. They proposed a simplified formulation based on a modified logarithmic law:

$$u^+ = \frac{1}{\psi\kappa} \ln(y^+) + C, \quad (2.36)$$

where ψ accounts for nonequilibrium effects. For equilibrium flows, $\psi = 1$, and the standard logarithmic law is recovered. They also combined this generalized wall function with the mixing function proposed by Kader [73] to extend its validity throughout the inner region.

Kalitzin et al. [81] introduced a lookup-table wall function based on well-resolved Couette-flow simulations. In this approach, nondimensional profiles of velocity, temperature, and turbulence variables are precomputed and stored for later use.

Temperature Wall Functions

As for velocity, the standard temperature wall function given in Equation 2.35 provides a simple representation of the near-wall thermal field. However, the buffer-layer behavior is difficult to describe using simple analytical expressions. Several temperature wall functions have therefore been proposed to provide continuous profiles across the inner region. Table 2.4 summarizes representative models.

The wall functions proposed by Craft et al. [78] and Popovac and Hanjalic [80] include heat-transfer effects in addition to velocity profiles. Another strategy is to use lookup tables for isothermal walls, where variations of T^+ with y^+ are precomputed. These approaches have also been extended to include pressure-gradient effects, as shown by Lacombe et al. [82].

Influence of Roughness

Surface roughness plays an important role in aircraft icing simulations. Roughness modifies the near-wall turbulence and can increase convective heat transfer, which directly affects the predicted ice shape. This is particularly important for glaze ice, where not all impinging water freezes immediately upon impact. Instead, a liquid film may run back from the stagnation region and freeze downstream, contributing to horn formation. In such cases, roughness is related to beads, rivulets, and irregular surface features that develop during ice accretion [85].

The pioneering experiments of Nikuradse [86] on sand-roughened pipes provided fundamental insight into rough-wall turbulent flows. One of the main observations was that roughness produces a downward shift of the logarithmic velocity profile, as illustrated in Figure 2.12. The rough-wall logarithmic law can be written as

$$u^+ = \frac{1}{\kappa} \ln(y^+) + C - \Delta u^+, \quad (2.37)$$

Table 2.4 Temperature Wall Functions

Source	SA or $k - \omega$ SST	Temperature Profile
Kader [73]	Both	$\theta^+ = Pr y^+ e^{-\tau} + 2.12 \ln((1 + y^+) + \beta(Pr)) e^{-\frac{1}{\tau}}$ $\tau = \frac{0.01(Pr y^+)^4}{1 + 5Pr^3 y^+}, \beta = (3.85Pr^{\frac{1}{3}} - 1.3)^2 + 2.12 \ln(Pr)$
Jayatillaka [83]	Both	$T^+ = Pr_t(u^+ + P_f)$ $P_f = 0.24[(\frac{Pr}{Pr_t})^{\frac{3}{4}} - 1][1 + 0.28 \exp(-0.007 \frac{Pr_t}{Pr})]$
Arpaci [84]	Both	$T^+ = \begin{cases} Pr y^+ & y^+ < y_1^+ \\ a_2 - \frac{Pr_t}{2a_1(y^+)^2} & y_1^+ \leq y^+ < y_2^+ \\ \frac{Pr_t}{\kappa} \ln(y^+) + \beta & y^+ \geq y_2^+ \end{cases}$

where Δu^+ is the roughness function.

Nikuradse classified roughness effects using the nondimensional roughness height. Schlichting [87] later showed that a single geometric parameter is generally insufficient to describe the effect of arbitrary roughness elements. Roughness spacing, shape, inclination, and density can all influence the flow response. This led to the concept of equivalent sand-grain roughness k_s , defined as the roughness height that produces the same shift in the logarithmic velocity profile as the real surface. This parameter is widely used in modern CFD solvers.

Several wall-function extensions include roughness effects by introducing a shift in the velocity profile. As summarized by Prakash et al. [9], the rough-wall velocity profile can be related to the smooth-wall profile through

$$\frac{du_{rough}^+}{dy^+}(y^+) = \frac{du_{smooth}^+}{dy^+}(y^+ + y_0^+), \quad (2.38)$$

$$u_{rough}^+(y^+) = u_{smooth}^+(y^+ + y_0^+) - u_{smooth}^+(y_0^+), \quad (2.39)$$

and

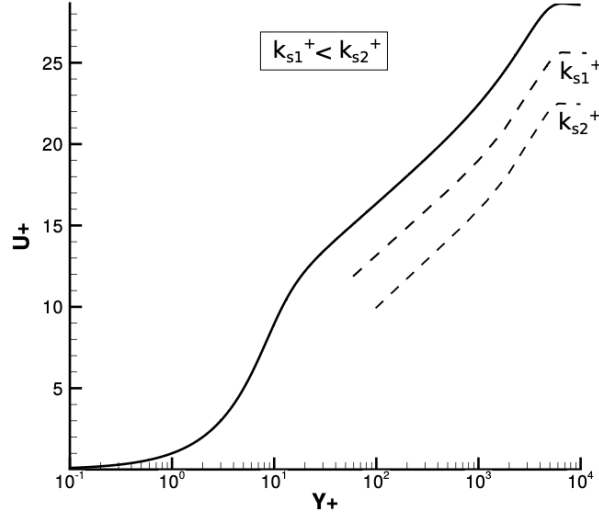


Figure 2.12 Influence of roughness on the velocity profile [9].

$$\Delta u^+ = u_{smooth}^+(y_0^+). \quad (2.40)$$

Here, y_0^+ is a wall shift related to the nondimensional roughness height k_s^+ . Similar corrections can be applied to temperature wall functions.

The influence of roughness is commonly divided into three regimes:

- **Hydraulically smooth regime** ($k_s^+ < 5$): the roughness elements are contained within the viscous sublayer, and their effect on skin friction is small. The surface can be treated as smooth;
- **Transitionally rough regime** ($5 \leq k_s^+ \leq 70$): roughness begins to affect the near-wall flow, the viscous sublayer is reduced, and skin friction increases. This regime is sensitive to roughness shape, spacing, and distribution;
- **Fully rough regime** ($k_s^+ > 70$): the viscous sublayer is disrupted, and the skin-friction coefficient becomes largely independent of Reynolds number. The flow response is mainly controlled by the roughness geometry.

In aircraft icing simulations, roughness modeling remains a significant source of uncertainty. Since roughness affects heat transfer and water runback, it can strongly influence glaze-ice accretion and horn formation. A robust numerical framework must therefore account for roughness effects consistently within the flow, heat-transfer, and thermodynamic models.

**Article I: A Ray-Tracing-Enhanced Lagrangian Particle Tracking Framework
for Ice Accretion Applications**

CHAPTER 3 CONCLUSION

3.1 Summary of Works

3.2 Limitations

3.3 Future Research

Texte / Text.

REFERENCES

- [1] “CRM-HL | NASA Common Research Model,” Apr. 2022. [Online]. Available: <https://commonresearchmodel.larc.nasa.gov/high-lift-crm/>
- [2] L. Xie, P. Li, H. Chen, and H. Liu, “Robust and efficient prediction of the collection efficiency in icing accretion simulation for 3d complex geometries using the lagrangian approach i: an adaptive interpolation method based on the restricted radial basis functions,” *International Journal of Heat and Mass Transfer*, vol. 150, p. 119290, 2020.
- [3] E. Whalen, A. Broeren, M. Bragg, and S. Lee, “Characteristics of runback ice accretions on airfoils and their aerodynamics effects,” in *43rd AIAA Aerospace Sciences Meeting and Exhibit*, 2005, p. 1065.
- [4] P. Ball, “What Makes a Droplet Splash?” *Physics*, vol. 7, p. 73, Jul. 2014, publisher: American Physical Society. [Online]. Available: <https://physics.aps.org/articles/v7/73>
- [5] D. K. Clarke, M. Salas, and H. Hassan, “Euler calculations for multielement airfoils using cartesian grids,” *AIAA journal*, vol. 24, no. 3, pp. 353–358, 1986.
- [6] R. Mittal, H. Dong, M. Bozkurttas, F. Najjar, A. Vargas, and A. Von Loebbecke, “A versatile sharp interface immersed boundary method for incompressible flows with complex boundaries,” *Journal of computational physics*, vol. 227, no. 10, pp. 4825–4852, 2008.
- [7] F. Capizzano and E. Iuliano, “A eulerian method for water droplet impingement by means of an immersed boundary technique,” *Journal of Fluids Engineering*, vol. 136, no. 4, p. 040906, 2014.
- [8] R. Scardovelli and S. Zaleski, “Direct numerical simulation of free-surface and interfacial flow,” *Annual review of fluid mechanics*, vol. 31, no. 1, pp. 567–603, 1999.
- [9] A. Prakash, “Rans computation of heat transfer over rough surfaces,” Ph.D. dissertation, Ecole Polytechnique, Montreal (Canada), 2019.
- [10] W. M. Leary, *We Freeze to Please: A History of NASA’s Icing Research Tunnel and the Quest for Flight Safety*. National Aeronautics and Space Administration, 2002, vol. 4226.

- [11] F. T. Lynch and A. Khodadoust, “Effects of ice accretions on aircraft aerodynamics,” *Progress in Aerospace Sciences*, vol. 37, no. 8, pp. 669–767, 2001.
- [12] Y. Cao, Z. Wu, Y. Su, and Z. Xu, “Aircraft flight characteristics in icing conditions,” *Progress in Aerospace Sciences*, vol. 74, pp. 62–80, 2015.
- [13] M. G. Potapczuk, “Aircraft icing research at nasa glenn research center,” *Journal of Aerospace Engineering*, vol. 26, no. 2, pp. 260–276, 2013.
- [14] “In-Flight Icing: Flight Operations - Departure - Act.” [Online]. Available: https://aircrafticing.grc.nasa.gov/1_2_4_2.html
- [15] M. G. Potapczuk and J. Reinmann, “Icing simulation: A survey of computer models and experimental facilities,” in *Fluid Dynamics Panel Specialists Meeting*, no. E-6164, 1991.
- [16] R. K. Jeck, “Icing design envelopes (14 cfr parts 25 and 29, appendix c) converted to a distance-based format,” Federal Aviation Administration, Tech. Rep. DOT/FAA/AR-00/30, 2002.
- [17] M. Oleskiw, “A review of 65 years of aircraft in-flight icing research at nrc,” *Canadian aeronautics and space journal. Le journal aéronautique et spatial du Canada*, vol. 47, 09 2001.
- [18] H. Ferschitz, M. Wannemacher, O. Bucek, F. Knöbel, and W. Breitfuß, “Development of sld capabilities in the rta icing wind tunnel,” *SAE International Journal of Aerospace*, vol. 10, no. 2017-01-9001, pp. 12–21, 2017.
- [19] P. Berthoumieu, B. Déjean, V. Bodoc, and T. Alary, *ONERA Research Icing Wind Tunnel*.
- [20] A. P. Broeren, M. Potapczuk, J. Riley, P. Villedieu, F. Moens, and M. Bragg, *Swept-Wing Ice Accretion Characterization and Aerodynamics*.
- [21] P. Lavoie, S. Bourgault-Côté, and E. Laurendeau, “Numerical algorithms for infinite swept wing ice accretion,” *Computers & Fluids*, vol. 161, pp. 189–198, 2018.
- [22] E. Laurendeau, S. Bourgault-Cote, I. A. Ozcer, R. Hann, E. Radenac, and A. Pueyo, “Summary from the 1st aiaa ice prediction workshop,” in *AIAA Aviation 2022 Forum*, 2022, p. 3398.

- [23] E. Laurendeau, M. Blanchet, M. K. Zayni, R. Hann, E. Radenac, I. Mussa, and A. Pueyo, “Summary from the 2nd aiaa ice prediction workshop,” in *AIAA AVIATION FORUM AND ASCEND 2024*. American Institute of Aeronautics and Astronautics, Jul. 2024.
- [24] D. Pena, Y. Hoarau, and E. Laurendeau, “A single step ice accretion model using level-set method,” *Journal of Fluids and Structures*, vol. 65, pp. 278–294, 2016.
- [25] S. Bourgault-Côté, J. Docampo-Sánchez, and E. Laurendeau, “Multilayer Airfoil Ice Accretion Simulations Using a Level-Set Method with B-Spline Representation,” *AIAA Journal*, vol. 57, no. 8, pp. 3299–3308, Aug. 2019.
- [26] P. Lavoie, E. Radenac, G. Blanchard, E. Laurendeau, and P. Villedieu, “Immersed boundary methodology for multistep ice accretion using a level set,” *Journal of Aircraft*, vol. 59, no. 4, pp. 912–926, 2022.
- [27] M. Blanchet, S. Bourgault-Cote, and E. Laurendeau, “Conservative hyperbolic droplet solver for aircraft icing,” in *AIAA AVIATION 2022 Forum*, 2022, p. 3369.
- [28] Y. Bourgault, W. G. Habashi, J. Dompierre, and G. S. Baruzzi, “A finite element method study of eulerian droplets impingement models,” *International Journal for Numerical methods in fluids*, vol. 29, no. 4, pp. 429–449, 1999.
- [29] D. L. Darmofal and R. Haimes, “An analysis of 3d particle path integration algorithms,” *Journal of Computational Physics*, vol. 123, no. 1, pp. 182–195, 1996.
- [30] T. Bellosta, G. Baldan, G. Sirianni, and A. Guardone, “Lagrangian and eulerian algorithms for water droplets in in-flight ice accretion,” *Journal of Computational and Applied Mathematics*, vol. 429, p. 115230, 2023.
- [31] Z. Li, Y. Wang, and L. Wang, “A fast particle-locating method for the arbitrary polyhedral mesh,” *Algorithms*, vol. 12, no. 9, p. 179, 2019.
- [32] T. Rendall and C. B. Allen, “Finite-volume droplet trajectories for icing simulation,” *International journal of multiphase flow*, vol. 58, pp. 185–194, 2014.
- [33] M. Papadakis, K. Hung, H.-W. Yeong, C. Bidwell, and M. Breer, “Experimental investigation of water impingement on single and multi-element airfoils,” in *38th Aerospace Sciences Meeting and Exhibit*, 1999, p. 100.
- [34] M. Papadakis, A. Rachman, S.-C. Wong, H.-W. Yeong, K. Hung, and C. Bidwell, “Water impingement experiments on a naca 23012 airfoil with simulated glaze ice shapes,” in *42nd AIAA aerospace sciences meeting and exhibit*, 2004, p. 565.

- [35] M. Papadakis, S.-C. Wong, A. Rachman, K. E. Hung, G. T. Vu, and C. S. Bidwell, “Large and small droplet impingement data on airfoils and two simulated ice shapes,” Tech. Rep., 2007.
- [36] M. Papadakis, K. E. Hung, G. T. Vu, H. W. Yeong, C. S. Bidwell, M. D. Breer, and T. J. Bencic, “Experimental investigation of water droplet impingement on airfoils, finite wings, and an s-duct engine inlet,” Tech. Rep., 2002.
- [37] M. Papadakis, A. Rachman, S.-C. Wong, C. Bidwell, and T. Bencic, “An experimental investigation of sld impingement on airfoils and simulated ice shapes,” SAE Technical Paper, Tech. Rep., 2003.
- [38] C. Tan, M. Papadakis, D. Miller, T. Bencic, P. Tate, and M. Laun, “Experimental study of large droplet splashing and breakup,” in *45th AIAA Aerospace Sciences Meeting and Exhibit*, 2007, p. 904.
- [39] M. Vargas and A. Feo, “Deformation and breakup of water droplets near an airfoil leading edge,” *Journal of aircraft*, vol. 48, no. 5, pp. 1749–1765, 2011.
- [40] P. Berthoumieu, “Experimental study of supercooled large droplets impact in an icing wind tunnel,” in *4th AIAA Atmospheric and Space Environments Conference*, 2012, p. 3130.
- [41] W. Wright and M. Potapczuk, “Semi-empirical modelling of sld physics,” in *42nd AIAA aerospace sciences meeting and exhibit*, 2004, p. 412.
- [42] C. Bhargava, E. Loth, and M. Potapczuk, “Numerical simulation of icing clouds in the nasa glenn icing research tunnel,” *Journal of aircraft*, vol. 42, no. 6, pp. 1442–1451, 2005.
- [43] P. Trontin and P. Villedieu, “A revisited model for sld impact onto a solid surface,” in *8th AIAA Atmospheric and Space Environments Conference*, 2016, p. 3277.
- [44] S. Guido, “Shear-induced droplet deformation: Effects of confined geometry and viscoelasticity,” *Current opinion in colloid & interface science*, vol. 16, no. 1, pp. 61–70, 2011.
- [45] Z. Xu, T. Wang, and Z. Che, “Droplet breakup in airflow with strong shear effect,” *Journal of Fluid Mechanics*, vol. 941, p. A54, 2022.
- [46] G. Cossali, G. Brunello, A. Coghe, and M. Marengo, “Impact of a single drop on a liquid film: experimental analysis and comparison with empirical models,” in *Italian congress of thermofluid dynamics UIT, Ferrara*, vol. 30, 1999, pp. 1–12.

- [47] M. Rein, “Phenomena of liquid drop impact on solid and liquid surfaces,” *Fluid dynamics research*, vol. 12, no. 2, pp. 61–93, 1993.
- [48] J. Alzaili and D. Hammond, “Experimental investigation of thin water film stability and its characteristics in sld icing problem,” SAE Technical Paper, Tech. Rep., 2011.
- [49] J. F. VanZante, “A database of supercooled large droplet ice accretions,” in *Aircraft and Engine Icing International Conference*, no. NASA/CR-2007-215020, 2007.
- [50] P. Trontin, G. Blanchard, A. Kontogiannis, and P. Villedieu, “Description and assessment of the new onera 2d icing suite igloo2d,” in *9th AIAA Atmospheric and Space Environments Conference*, 2017, p. 3417.
- [51] C. W. Hirt and B. D. Nichols, “Volume of fluid (vof) method for the dynamics of free boundaries,” *Journal of computational physics*, vol. 39, no. 1, pp. 201–225, 1981.
- [52] Y. Lian, “Numerical simulation of supercooled large droplets using the moment of fluid method,” in *52nd Aerospace Sciences Meeting*, 2014, p. 0740.
- [53] D. R. Bilodeau, W. G. Habashi, G. S. Baruzzi, and M. Fossati, “Numerical modeling of first and second order sld effects on 3d geometries,” in *8th AIAA Atmospheric and Space Environments Conference*, 2016, p. 3280.
- [54] F. Capizzano and D. de Rosa, “Extending the impingement capabilities of a cartesian solver towards super-cooled large droplets (sld),” SAE Technical Paper, Tech. Rep., 2023.
- [55] C. Mundo, M. Sommerfeld, and C. Tropea, “On the modeling of liquid sprays impinging on surfaces,” *Atomization and sprays*, vol. 8, no. 6, 1998.
- [56] M. Trujillo, W. Mathews, C. Lee, and J. Peters, “Modelling and experiment of impingement and atomization of a liquid spray on a wall,” *International journal of engine research*, vol. 1, no. 1, pp. 87–105, 2000.
- [57] A. L. Yarin and D. A. Weiss, “Impact of drops on solid surfaces: self-similar capillary waves, and splashing as a new type of kinematic discontinuity,” *Journal of fluid mechanics*, vol. 283, pp. 141–173, 1995.
- [58] R. Honsek and W. G. Habashi, “Fensap-ice: Eulerian modeling of droplet impingement in the sld regime of aircraft icing,” in *44th AIAA aerospace sciences meeting and exhibit*, 2006, p. 465.

- [59] F. Sotiropoulos and X. Yang, “Immersed boundary methods for simulating fluid–structure interaction,” *Progress in Aerospace Sciences*, vol. 65, pp. 1–21, 2014.
- [60] J. Glimm, X. Li, Y. Liu, Z. Xu, and N. Zhao, “Conservative front tracking with improved accuracy,” *SIAM Journal on Numerical Analysis*, vol. 41, no. 5, pp. 1926–1947, 2003.
- [61] M. Kirkpatrick, S. Armfield, and J. Kent, “A representation of curved boundaries for the solution of the navier–stokes equations on a staggered three-dimensional cartesian grid,” *Journal of Computational Physics*, vol. 184, no. 1, pp. 1–36, 2003.
- [62] M. Meyer, A. Devesa, S. Hickel, X. Y. Hu, and N. A. Adams, “A conservative immersed interface method for large-eddy simulation of incompressible flows,” *Journal of Computational Physics*, vol. 229, no. 18, pp. 6300–6317, 2010.
- [63] C. S. Peskin, “Flow patterns around heart valves: a numerical method,” *Journal of computational physics*, vol. 10, no. 2, pp. 252–271, 1972.
- [64] M.-C. Lai and C. S. Peskin, “An immersed boundary method with formal second-order accuracy and reduced numerical viscosity,” *Journal of computational Physics*, vol. 160, no. 2, pp. 705–719, 2000.
- [65] E. Arquis, J. Caltagirone *et al.*, “Sur les conditions hydrodynamiques au voisinage d’une interface milieu fluide-milieu poreux: application la convection naturelle,” *CR Acad. Sci. Paris II*, vol. 299, pp. 1–4, 1984.
- [66] J. Mohd-Yusof, “For simulations of flow in complex geometries,” *Annual research briefs*, vol. 317, p. 35, 1997.
- [67] Y.-H. Tseng and J. H. Ferziger, “A ghost-cell immersed boundary method for flow in complex geometry,” *Journal of computational physics*, vol. 192, no. 2, pp. 593–623, 2003.
- [68] F. Capizzano, “A compressible flow simulation system based on cartesian grids with anisotropic refinements,” in *45th AIAA Aerospace Sciences Meeting and Exhibit*, 2007, p. 1450.
- [69] —, “Turbulent wall model for immersed boundary methods,” *AIAA journal*, vol. 49, no. 11, pp. 2367–2381, 2011.
- [70] J. A. Sethian and P. Smereka, “Level set methods for fluid interfaces,” *Annual review of fluid mechanics*, vol. 35, no. 1, pp. 341–372, 2003.

- [71] P. Rauschenberger, A. Criscione, K. Eisenschmidt, D. Kintea, S. Jakirlić, Ž. Tuković, I. Roisman, B. Weigand, and C. Tropea, “Comparative assessment of volume-of-fluid and level-set methods by relevance to dendritic ice growth in supercooled water,” *Computers & fluids*, vol. 79, pp. 44–52, 2013.
- [72] I. Marter, “Méthode d’interface immergée pour la simulation directe de l’atomisation primaire,” Ph.D. dissertation, Institut supérieur de l’Aéronautique et de l’Espace (ISAE); Université de . . . , 2017.
- [73] B. Kader and A. Yaglom, “Heat and mass transfer laws for fully turbulent wall flows,” *International Journal of Heat and Mass Transfer*, vol. 15, no. 12, pp. 2329–2351, 1972.
- [74] D. B. Spalding *et al.*, “A single formula for the law of the wall,” *Journal of Applied Mechanics*, vol. 28, no. 3, pp. 455–458, 1961.
- [75] H. Reichardt, “Vollständige darstellung der turbulenten geschwindigkeitsverteilung in glatten leitungen,” *ZAMM-Journal of Applied Mathematics and Mechanics/Zeitschrift für Angewandte Mathematik und Mechanik*, vol. 31, no. 7, pp. 208–219, 1951.
- [76] T. Knopp, T. Alrutz, and D. Schwamborn, “A grid and flow adaptive wall-function method for rans turbulence modelling,” *Journal of Computational Physics*, vol. 220, no. 1, pp. 19–40, 2006.
- [77] S. R. Allmaras and F. T. Johnson, “Modifications and clarifications for the implementation of the spalart-allmaras turbulence model,” in *Seventh international conference on computational fluid dynamics (ICCFD7)*, vol. 1902. Big Island, HI, 2012.
- [78] T. Craft, A. Gerasimov, H. Iacovides, and B. Launder, “Progress in the generalization of wall-function treatments,” *International Journal of Heat and Fluid Flow*, vol. 23, no. 2, pp. 148–160, 2002.
- [79] S. E. Gant, *Development and application of a new wall function for complex turbulent flows*. The University of Manchester (United Kingdom), 2002.
- [80] M. Popovac and K. Hanjalic, “Compound wall treatment for rans computation of complex turbulent flows and heat transfer,” *Flow, turbulence and combustion*, vol. 78, pp. 177–202, 2007.
- [81] G. Kalitzin, G. Medic, G. Iaccarino, and P. Durbin, “Near-wall behavior of rans turbulence models and implications for wall functions,” *Journal of Computational Physics*, vol. 204, no. 1, pp. 265–291, 2005.

- [82] F. Lacombe, D. Pelletier, and A. Garon, “Compatible wall functions and adaptive remeshing for the k-omega sst model,” in *AIAA Scitech 2019 Forum*, 2019, p. 2329.
- [83] C. L. V. Jayatilleke, “The influence of prandtl number and surface roughness on the resistance of the laminar sub-layer to momentum and heat transfer,” 1966.
- [84] V. Arpaci and P. Larsen, “Heat engineering convection heat transfer,” 1984.
- [85] R. J. Hansman Jr and S. R. Turnock, “Investigation of surface water behavior during glaze ice accretion,” *Journal of Aircraft*, vol. 26, no. 2, pp. 140–147, 1989.
- [86] L. Nikuradse, “Laws of flow in rough pipes, vdi forsch. 361,” *English translation. NACA TM-1292*, 1933.
- [87] H. Schlichting, “Experimentelle untersuchungen zum rauigkeitsproblem,” *Ingenieur-Archiv*, vol. 7, no. 1, pp. 1–34, 1936.

APPENDIX A DÉMO

Texte de l'annexe A. Remarquez que la phrase précédente se termine par une lettre majuscule suivie d'un point. On indique explicitement cette situation à $\text{\LaTeX}$ afin que ce dernier ajuste correctement l'espacement entre le point final de la phrase et le début de la phrase suivante.

APPENDIX B UNE DERNIÈRE ANNEXE / THE LAST APPENDIX

Texte de l'annexe C.